

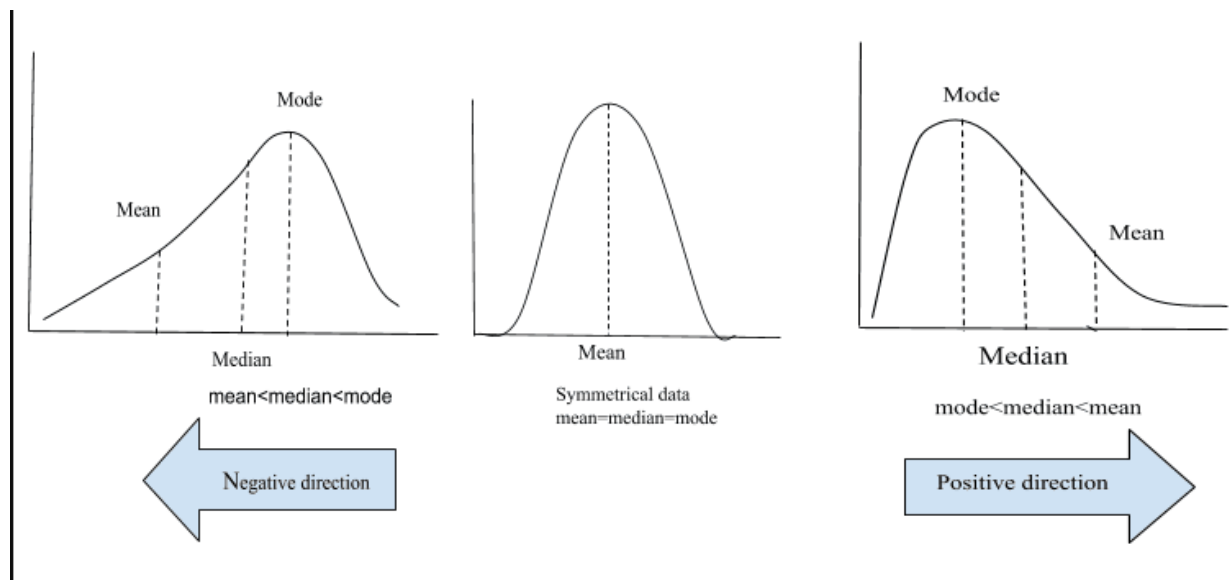
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**Title: Data transformation to handle skewness in
machine learning**

1. Introduction:

skewness is a statistical measure that describes how asymmetry of a probability distribution around its mean, it basically tells us which direction our data is leaning towards.

2. Understanding skewness:



Type	Tail direction	Mean,mode,median relationship	Real-world example
Symmetric	Equal on both sides	Mean = median = mode	Human height in a population
+ve skew	Extends to the right	Mean > median > mode	House hold prices
-ve skew	Extends to the left	Mean < median < mode	Age at death

Skewness tells us how our data is leaning, it gives us a sense of what values are dominating our data.

When skewness is positive (to the right), most of the data is concentrated on the lower end with outliers on the higher end effectively pushing the mean to the right,

and when skewness is negative (to the left), most of the data is concentrated on the higher end with outliers on the lower end effectively pulling the mean to the left.

It falls under the formula $\gamma_1 = E \left[\left(\frac{x - \mu}{\sigma} \right)^3 \right]$.

Where $x - \mu$ is the distance from mean and the σ is the standard deviation

3. **Why can skewness be harmful:**

3.1 distorts statistics:

When data is skewed and the mean gets pulled towards the tail, filling values using the mean is inaccurate as the mean is higher/lower than most of the data introducing bias to the data, also standard deviation and variance rely on the mean so any distortion in it distorts the standard deviation and thus the z-score

3.2 Warps Distance-Based Algorithms:

Algorithms that compute distances between points are heavily sensitive to extreme tails

knn and k-means models :

Extreme tail values dominate distance calculations. This forces clusters to stretch toward outliers or causes misclassification of normal points.

3.3 Violates Assumptions of Linear & Parametric Models:

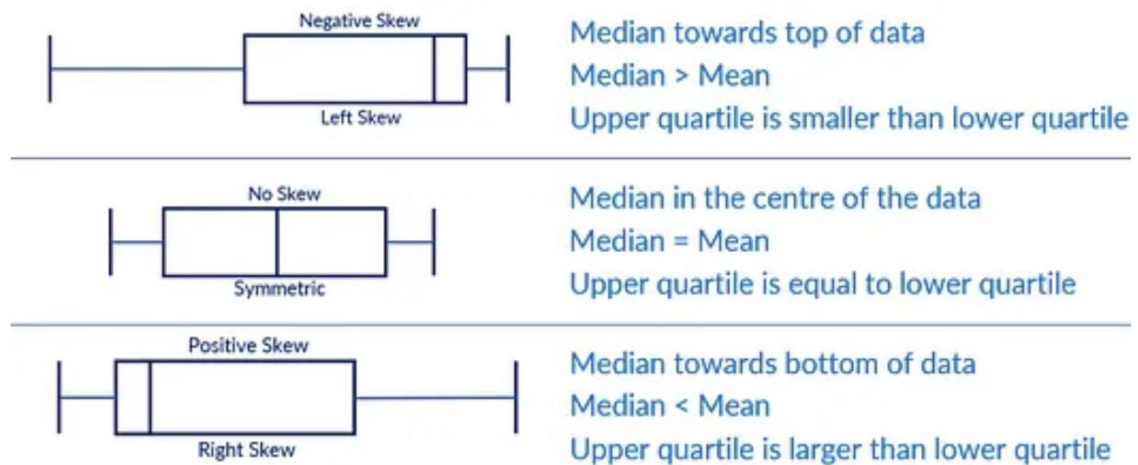
Many classic algorithms assume features or residual errors follow a normal distribution

Linear and logistic regression:

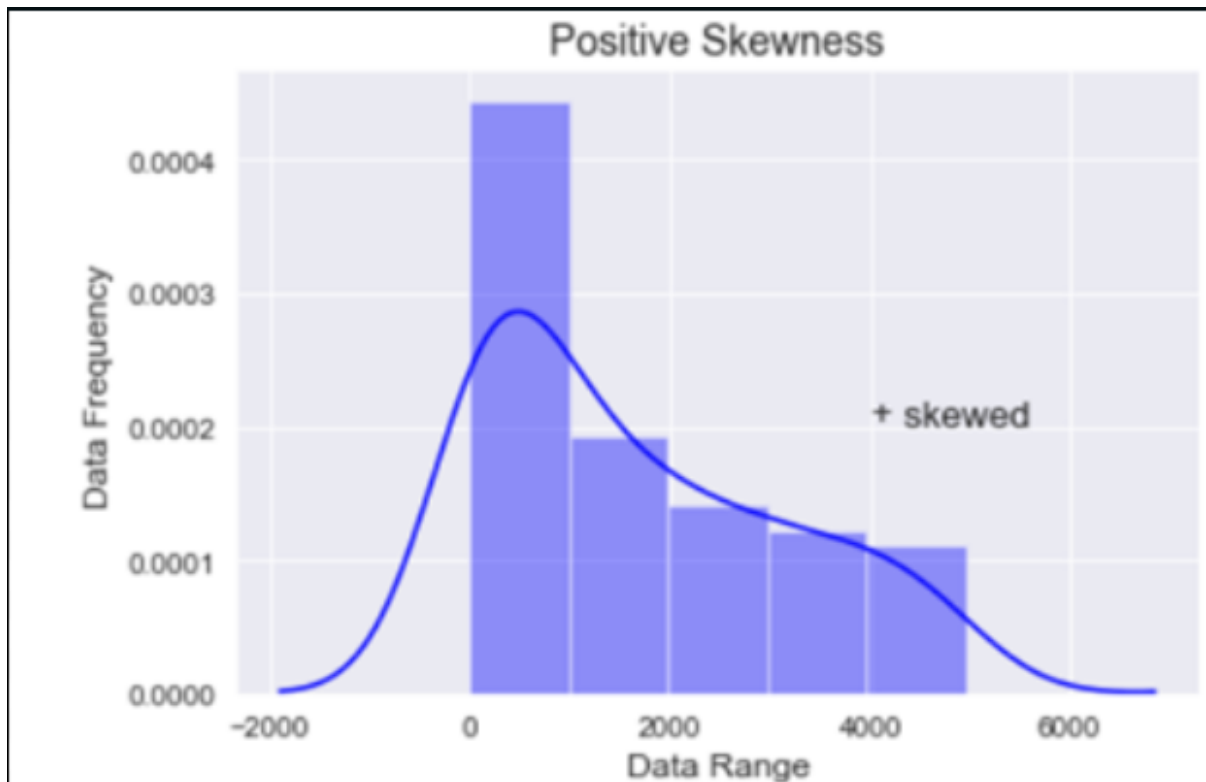
due to the skewness, the model makes far larger prediction errors on extreme tail values than on typical values.

4. Detecting skewness:

There are a couple of ways to detect skewness such as box plots which depend on IQR:



and histograms with KDE (Kernel density estimation):



the blue line is the KDE and its basically an estimation of how the distribution curve looks for a certain data

5. Handling positive skewness:

We already know that we must handle skewness for gradient/distance-based models to work correctly different methods have been to compress the tail and bring the data close to a normal distribution

5.1 Log transformation

Compresses higher values much more than lower values, it's best for severe right skewness spanning several orders of magnitude, it cannot handle negative values.

5.2 Square root transformation

Applies mild compression to high values, best for mild data and cannot handle negative values.

5.3 Cubic root transformation

Similar to a square root but slightly stronger compression and can handle negative values naturally, works best with moderate data that has negative values without needing a constant shift.

5.4 Reciprocal transformation

Inverts the scale, small numbers become extremely large while large numbers become extremely small, it has drawbacks where it reverses the whole order and cannot handle 0 values.

5.5 box-cox transformation

$$y^{(\lambda)} = \begin{cases} \frac{y^\lambda - 1}{\lambda}, & y \neq 0 \\ \ln(y), & y = 0 \end{cases}$$

It picks lambda for positive values of our data and an optimization algorithm (most likelihood estimation) estimates a value for lambda that would transform and yield the maximum normalization, it cannot handle negative values

5.6 Yeo Johnson

The most widely used algorithm for handling skew data, can handle negative, positive and 0 values, it's an extension for box-cox

6. Handling negative skewness

Negative skewness happens when a dataset has a long tail on the left side or extending towards smaller numbers with most values concentrated on the right, standard algorithms like box-cox are designed to fit right skewness by compressing large values so using them on negative skewness just makes the problem worse.

Instead of creating a new algorithm specially for negative data, the dataset is reflected or mirrored along the y axis. Once reflected, the data is considered a positive skew, and the algorithms will work.

Yeo Jhonson can handle negative data without the need for mirroring or flipping as it can handle negative values and its generally preferred as it doesn't need further investigating.

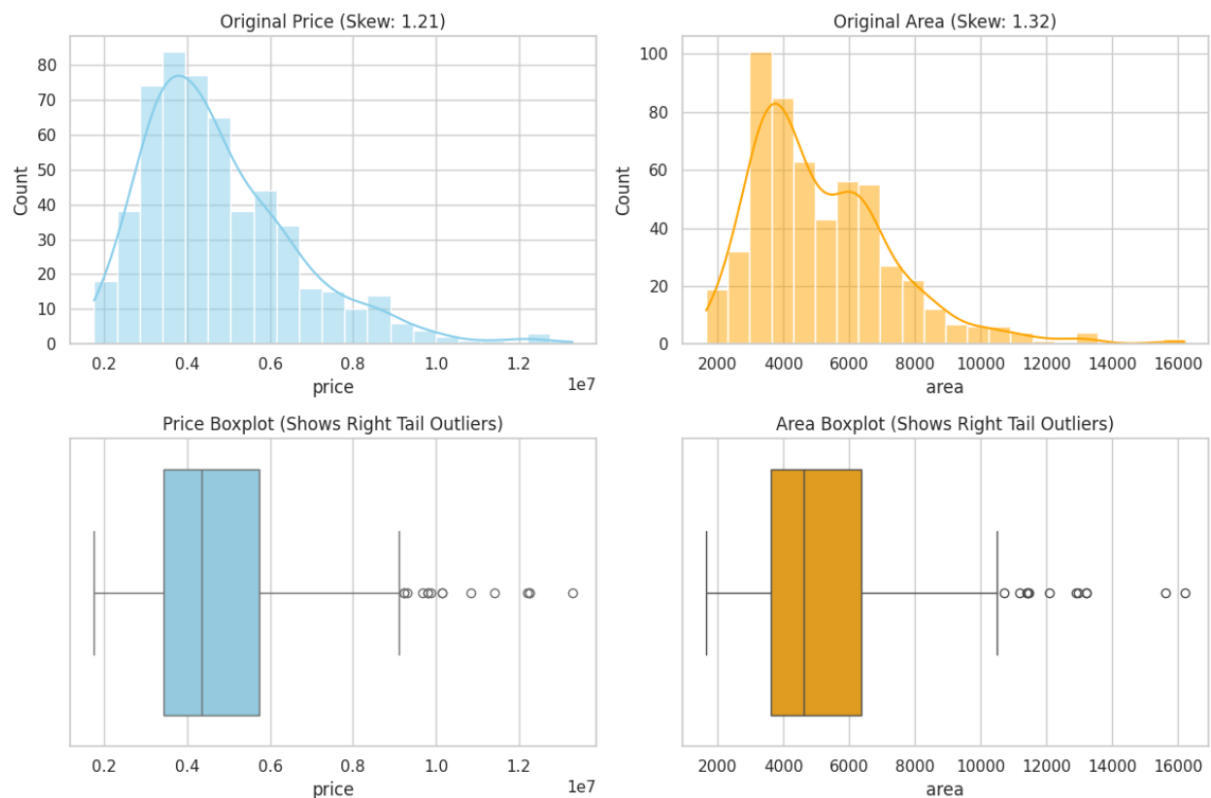
7. Box-cox vs Yeo Jhonson comparison

Feature / Metric	Box-Cox	Yeo-Johnson
Formula	$y^{(\lambda)} = \begin{cases} \frac{y^\lambda - 1}{\lambda}, & \text{if } \lambda \neq 0 \\ \ln(y), & \text{if } \lambda = 0 \end{cases}$	$y^{(\lambda)} = \begin{cases} \frac{(y+1)^\lambda - 1}{\lambda}, & \text{if } \lambda \neq 0 \text{ and } y \geq 0 \\ \ln(y + 1), & \text{if } \lambda = 0 \text{ and } y \geq 0 \\ -\frac{(-y+1)^{2-\lambda} - 1}{2-\lambda}, & \text{if } \lambda \neq 2 \text{ and } y < 0 \\ -\ln(-y + 1), & \text{if } \lambda = 2 \text{ and } y < 0 \end{cases}$ w
Data type	Only works with positive data	Works with all types of data
Advantages	Mathematically simpler Highly effective for right-skewed positive datasets Straightforward parameter estimation and inversion.	Handles zeros and negative numbers without constant offsets or reflections More versatile for modern machine learning pipelines where features are standardized or centered around zero.
Disadvantages	Needs a constant shift or offset to deal with negative values Requires reflection to handle left-skewed data.	slightly more complex math. can still be sensitive to extreme outliers, which may bias parameter estimation. Interpretability of transformed units remains abstract (requires inverse transformation for physical meaning).
Suitable Use Cases	positive physical measurement data	Normalized, mean-centered, or standardized datasets with mixed negative and positive values.

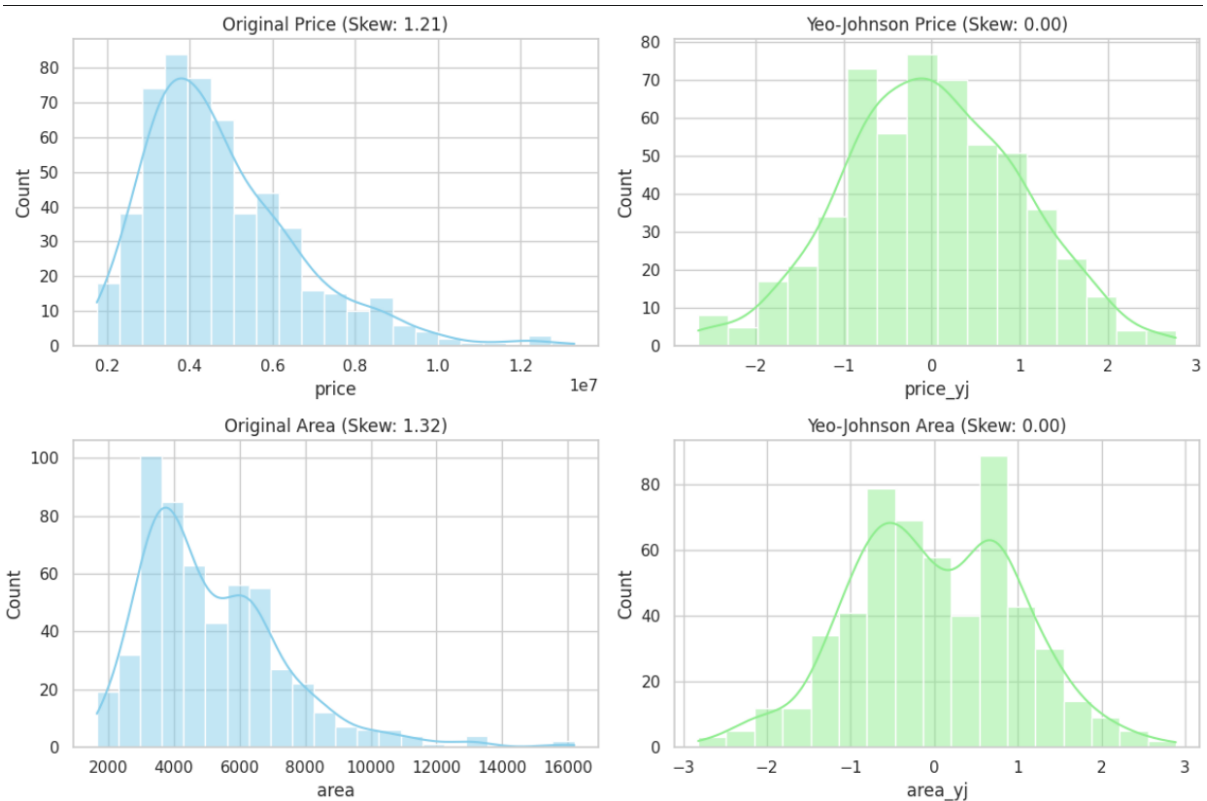
	Financial metrics bound above zero (house prices, income).	Datasets containing zeros Feature engineering in scikit-learn pipelines
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8. Practical example:

while exploring “housing.csv” file I found out most the data was right skewed



the prices and areas of the houses were positively skewed. We can see the outliers in the boxplots, so Yeo Johnson applied.



and these were the plots before and after and we can clearly see its much more normalized now.

9. Conclusion

Data transformation is vital for data processing and for ML models to work correctly, handling skewness stabilizes variance and neutralizes the effect of outliers and ensures that distance based and gradient models can work correctly, while traditional transformation functions and box-cox can work for certain data types, pipelines are leaning towards Yeo Jhonson due to its versatility.

10. References

- **Box-Cox Transformation:** Box, G. E. P., & Cox, D. R. (1964). An analysis of transformations. *Journal of the Royal Statistical Society: Series B (Methodological)*, 26(2), 211–243. <https://doi.org/10.1111/j.2517-6161.1964.tb00553.x>

- **Yeo-Johnson Transformation:** Yeo, I.-K., & Johnson, R. A. (2000). A new family of power transformations to improve normality or symmetry. *Biometrika*, 87(4), 954–959. <https://doi.org/10.1093/biomet/87.4.954>
- **General Machine Learning & Data Preprocessing:** Kuhn, M., & Johnson, K. (2013). *Applied Predictive Modeling*. Springer.